Where Do Text Embeddings Keep Geography? Layer-wise Probes and Map Views of Base vs. Retrieval-Tuned Models

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GitHub repository: `https://github.com/afrenkai/ComeALittleCloser`

Progress since the proposal

- **Probing pipeline (complete).** Ridge probes on every layer's pooled hidden state for 34,196 GeoNames cities and 11,833 radio stations; 10° spatial-block and random splits, 3 seeds, block-bootstrap CIs, a country-centroid control, untouched test split.
- **Models.** 18 checkpoints: five decoder *base→retrieval* pairs (Qwen3 0.6B/4B/8B; Mistral-7B→e5-mistral; Gemma-2-9B→bge-multilingual-gemma2), XLM-R base/large with E5 and the BGE-M3 stages, two random-init controls.
- **Validation results (all 18 models; test not yet run).** A *country-residual* probe (a place's offset from its country's centroid) finds within-country geography in every decoder from 4B up: at the best layer it beats the country centroid by 12 / 24, 120 / 152 and 175 / 201 km for Qwen3 0.6B / 4B / 8B, 248 / 304 for Mistral-7B and 303 / 285 for Gemma-2-9B (base / retrieval; cities); encoders stay near 0. Base and retrieval CIs overlap in all five pairs: retrieval training keeps this information. Absolute-location probes on the Qwen3-8B, e5-mistral and bge-multilingual-gemma2 retrieval models gain 311–459 km on cities, none on stations.
- **Retrieval.** A frozen run (5 models, 11,479 stations, 45 genres, 12 origins): changing the query origin replaces 8–21% of the top 10, yet regret against the true nearest stays at 3,668–4,141 km; adding the origin changes regret by −170 km (CLAP) to +385 km (EmbeddingGemma).
- **Visualization.** *Built:* the globe explorer (origin/genre selection, true vs. model top-k on a 3-D globe, audio), the basis of V1; static depth curves with CIs for all 18 models; a static true→predicted arrow map (Qwen3-8B). *Planned:* interactive V2, V3, linking.

Response to feedback

1. **Visualization contribution.** Three views share one selection (model pair, layer, region, place), so a pattern seen in one is checked in the others. *(V1) Globe ranking:* the model's top 10 and the true 10 nearest as arcs from the origin, colored same-country / same-language / other, so border stops and language jumps show; a panel gives the query's metrics against the random-ordering band. *(V2) Layer-by-region map:* 10° blocks colored by probe gain over the country centroid. Brushing a layer range on the depth curve updates the map; toggles switch model, probe target and base−retrieval difference; clicking a block shows its depth curve with CI, its places in V3, and sets V1's origin there. *Signal vs. noise:* a block is colored only with ≥30 places and a CI excluding 0 after multiple-comparison correction (else hatched); the random-init model gives a tokenization-only reference map. *(V3) Distortion map:* each block shows its mean displacement (predicted minus true, local km) with a bootstrap CI. To separate overlapping pulls, displacements are regressed jointly on unit vectors toward the capital, the country centroid and the continent centroid; blocks where these directions nearly coincide are marked not identifiable. Difference arrows show what retrieval training moves. *Evaluation (no IRB):* case studies C1–C4 below, each confirmed by a number, and a think-aloud walkthrough with 2–3 NLP/GIS researchers.
2. **Critique of key papers.** *Decoupes et al.* use final embeddings of isolated city names only, so geography never learned cannot be told apart from geography lost before the output; they measure distance correspondence, not whether distortions change search results. *Gurnee & Tegmark* probe base-LM layers with R^2 on raw latitude/longitude, which rewards knowing the country: on our data the country centroid alone gives $R^2 = 0.95$ (latitude), and a ridge probe on a frozen, randomly initialized Qwen3-0.6B reaches 0.78 when the text names the country. Probes show decodability, not use, and they cover generation models only. *Our fixes:* layer-wise probes, a country-residual target with both controls, retrieval metrics on the same places, base→retrieval pairs.
3. **Scope.** Core: layer probes (base vs. retrieval), retrieval metrics with baselines, border-crossing and coordinate-perturbation tests, V1–V3 with case studies. Optional: H3 adjacency, SoundingEarth audio with LLM re-captioning, multimodal comparison.
4. **Subspace amplification** is dropped. Coordinate perturbation stays; it shows whether rankings respond to coordinate *text*, not *why*.
5. **Metrics and framing.** Per (origin, genre) query over the genre-eligible pool: Spearman ρ between cosine and Haversine ranks (does the *whole* ranking follow distance?), overlap@10 with the 10 nearest (is the *top* a user sees right?), and excess mean km of the top 10 (*how far off* are the misses?). High ρ with low overlap means coarse order but wrong neighbors. Baselines: random ordering of the pool (1,000 permutations) and the Haversine sort. Framing: *which notion of proximity embeddings encode* (same-country and same-language as alternatives). **Decodable vs. reflected:** probe results are reported as *decodable*, retrieval metrics as *reflected in retrieval*. On the same stations, over blocks with ≥30 stations and ≥3 query origins, we report the Spearman correlation between final-layer probe gain and mean excess km. High gain with large excess is *consistent with* geography that is decodable but not reflected in retrieval; this is correlational, not proof the model ignores it.
6. **Small fixes.** New title; model count clarified (Jina dropped; EmbeddingGemma has no released base); typos fixed; SoundingEarth uses recorded coordinates.

Case studies and plan

Pre-stated case studies, each answered in a view and checked by a number: **C1** (V2): Is within-country geography uneven across regions? Spread of per-block gain. **C2** (V2 difference): Does retrieval training lose geography in some regions while the global mean holds? Significantly negative blocks. **C3** (V3): Are errors pulls toward capitals or continent centers? Joint regression coefficients with CIs. **C4** (V2→V1): Where geography is decodable, does retrieval reflect it? The correlation in 5 and border-crossing rate there.

- **Owners.** *Tian-hao Zhang:* probes, test run, per-block scores for V2/V3, coordinate perturbation. *Artem Frenk:* V1 and V2 interface, linking. *Nathan Chen:* retrieval metrics, baselines, ∼300 origins, border crossing. *Kenta Ploch:* V3, case studies, walkthrough.
- **Nov 4–10:** test cells frozen and test run (Zhang); V2 skeleton on validation scores (Frenk); metrics code (Chen); V3 aggregation (Ploch). **Nov 11–24:** per-block test scores into V2/V3; metrics on ∼300 origins, border crossing, perturbation; V2 and V3 linked to V1. **Nov 25–Dec 6:** C1–C4, walkthrough. **Dec 7–20:** paper. Fallback if V2 lags: static per-layer maps for C1–C2.
- **Status vs. proposal schedule.** Weeks 1 and 3 done (snapshots, manifests, city probes with controls); week 2 partly (outputs cached; lexical and fusion baselines with Chen); week 4, linked visualization, is behind: our main risk, hence the reassignment.